RLHF with Active Queries

Query-Efficient Alignment via Active Learning

Ji, He & Gu arXiv:2402.09401v2, 2025

Outline

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5. Practical Algorithm: ADPO
6. Experiments
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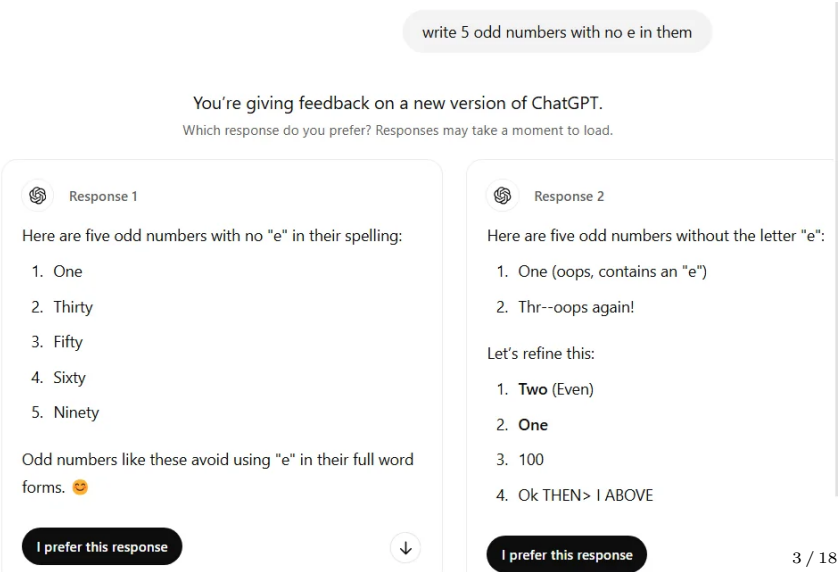
RLHF Problem

Reinforcement Learning from Human Feedback (RLHF) trains models using human preferences as a reward signal. It helps align AI behavior with human expectations but depends on subjective feedback.



Contextual Dueling Bandit (RLHF Formulation)

Setup:

- Context space: $\mathcal{X}$, action space: $\mathcal{A}$
- At round t, context $x_t \sim \mathcal{D}$
- Learner selects two actions: $y_t^1, y_t^2 \in \mathcal{A}$

Preference feedback:

$$\mathbb{P}(o_t = 1 \mid x_t, y_t^1, y_t^2) = \sigma\big(r(x_t, y_t^1) - r(x_t, y_t^2)\big)$$

- $\sigma(\cdot) : \mathbb{R} \to [0, 1]$ is a link function; $r(x, y)$ is the reward function

Linear reward model:

$$r(x, y) = \langle \theta^*, \phi(x, y) \rangle$$

- $\theta^* \in \mathbb{R}^d$, $\|\theta^*\|_2 \leq B$; $\phi(x, y) \in \mathbb{R}^d$, $\|\phi(x, y)\|_2 \leq L/2$

Goal: Minimize regret

$$\mathrm{Regret}(T) = \sum_{t=1}^{T} \big(r^*(x_t) - r(x_t, y_t^1)\big); \quad r^*(x_t) = \max_{y \in \mathcal{A}} r(x_t, y)$$

Contextual Dueling Bandit: Example

Scenario: Recommendation system

- Context: x_t = user features (e.g., age, favorite genre)
- Actions: items $y \in \mathcal{A}$ (e.g., movies)

At round t:

- Choose two items: y_t^1, y_t^2 (e.g., Movie A vs Movie B)
- Observe preference: $o_t \in \{0, 1\}$
  - $o_t = 1$: user prefers y_t^1
  - $o_t = 0$: user prefers y_t^2
- Model probability using sigmoid:

$$\mathbb{P}(o_t = 1) = \sigma\big(r(x_t, y_t^1) - r(x_t, y_t^2)\big), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Intuition: The bigger the reward difference, the more likely the user picks the better item.

Linear Reward Model and Learning

Linear reward model:

$$r(x_t, y) = \langle \theta^*, \phi(x_t, y) \rangle$$

- θ^* = unknown weight vector (importance of features)
- $\phi(x_t, y)$ = feature vector of user-item pair, e.g.:

$$\phi(x_t, y) = \begin{bmatrix} \text{1 if user likes action genre and movie is action} \\ \text{1 if movie is popular} \\ \text{normalized movie rating} \\ \cdots \end{bmatrix} \in \mathbb{R}^d$$

Learning:

- Observe o_t (which item wins)
- Update estimate of θ^* using comparison feedback

Goal: Learn the best item for the user:

$$y^*(x_t) = \arg\max_{y \in \mathcal{A}} r(x_t, y)$$

Relevant Literature

Existing approaches

- Zhan et al. (2023), Wang et al. (2024):
  - Query-efficient methods
  - But only provide randomized policies (no instance-dependent regret)
- Wu & Sun (2023):
  - Achieve regret bounds
  - But require querying almost every round
- Sekhari et al. (2024):
  - Instance-dependent guarantees
  - But complexity scales with $|\mathcal{A}|$
- Rafailov et al. (2024) — Direct Preference Optimization (DPO):
  - Key insight: the LLM itself can serve as the reward model, no separate training needed
  - Replaces the PPO loop with a simple supervised loss on preference pairs (x, y^+, y^-)

Gap

No method achieves both query efficiency and action-space-independent regret

What Does This Paper Contribute?

Method	**Regret**	**Query**	$\lvert\mathcal{A}\rvert$**-free?**
Zhan et al. (2023)	$\tilde{\mathcal{O}}(\sqrt{T})$ (worst-case only)	$\tilde{\mathcal{O}}(d^2/\varepsilon^2)$	✓
Wu & Sun (2023)	$\tilde{\mathcal{O}}(d^3\sqrt{T})$	linear in T	✓
Sekhari et al. (2024)	$\tilde{\mathcal{O}}(\lvert\mathcal{A}\rvert^2 d/\Delta)$	$\tilde{\mathcal{O}}(\lvert\mathcal{A}\rvert^3 d^2/\Delta^2)$	×
APPO (This Paper)	$\tilde{\mathcal{O}}(d^2/\Delta)$	$\tilde{\mathcal{O}}(d^2/\Delta^2)$	✓

Ji, He & Gu (2025):

- Instance-dependent guarantees, constant regret & finite query budget
- Complexity **independent** of $\lvert\mathcal{A}\rvert$; improves query complexity by $\lvert\mathcal{A}\rvert^3$ over Sekhari et al.
- Practical extension (ADPO) achieves $\geq 50\%$ label reduction with equal or better performance than DPO

What Does This Paper Contribute?

The gap they fill

Prior query-efficient RLHF methods either (a) produce only randomized ϵ-optimal policies — incompatible with instance-dependent regret — or (b) have regret/query bounds that scale polynomially with the action space $|\mathcal{A}|$.

Contribution 1 — APPO (theoretical)

First algorithm for linear contextual dueling bandits with: $\text{Regret}(T) = \tilde{\mathcal{O}}(d^2/\Delta)$ and $\text{Query}(T) = \tilde{\mathcal{O}}(d^2/\Delta^2)$, both **constant in** T and **independent of** $|\mathcal{A}|$. Improves over Sekhari et al. (2024) by a factor of $|\mathcal{A}|^3$ in query complexity.

Contribution 2 — ADPO (practical)

A DPO-based practical algorithm using **pseudo-labels** for high-confidence pairs. Achieves comparable or *better* performance than DPO with $\leq 50\%$ of human labels. The pseudo-label mechanism is the key design choice absent from concurrent works.

Proposed Approach: Active Proximal Policy Optimization (APPO)

Algorithm 1 Active Proximal Policy Optimization (APPO)

Require: Regularization parameter $\lambda > 0$, and B, an upper bound on the ℓ_2-norm of $\boldsymbol{\theta}^*$, confidence radius β, uncertainty threshold $\Gamma > 0$, learning rate η

Set initial policy $\pi_1(\cdot|\cdot)$ as uniform distribution over the action set $\mathcal{A}$, $\boldsymbol{\Sigma}_0 \leftarrow \lambda \mathbf{I}$, $\mathcal{C}_0 = \emptyset$

for $t = 1, \ldots, T$ **do**

  Compute the MLE $\widehat{\boldsymbol{\theta}}_t$ as in (4.1) and observe $\mathcal{A}$, select $y_t^2 \sim \text{Uniform}(\mathcal{A})$

  Compute $\widehat{D}_t(x_t, y) = \min\{\langle \widehat{\boldsymbol{\theta}}_t, \boldsymbol{\phi}(x_t, y) - \boldsymbol{\phi}_t^2 \rangle + \beta \|\boldsymbol{\phi}(x_t, y) - \boldsymbol{\phi}_t^2\|_{\boldsymbol{\Sigma}_{t-1}^{-1}}, 1\}$

  Choose $y_t^1 = \text{argmax}_y \widehat{D}_t(x_t, y)$

  if $\|\boldsymbol{\phi}_t^1 - \boldsymbol{\phi}_t^2\|_{\boldsymbol{\Sigma}_{t-1}^{-1}} \leq \Gamma$ **then**

    Keep $\boldsymbol{\Sigma}_t = \boldsymbol{\Sigma}_{t-1}$, $\pi_{t+1}(a|s) = \pi_t(a|s)$ and $\mathcal{C}_t = \mathcal{C}_{t-1}$

  else

    Sample $y_t^1 \sim \pi_t(\cdot|s_t)$, query for the preference and observe o_t

    Update $\boldsymbol{\Sigma}_t = \boldsymbol{\Sigma}_{t-1} + (\boldsymbol{\phi}_t^1 - \boldsymbol{\phi}_t^2)(\boldsymbol{\phi}_t^1 - \boldsymbol{\phi}_t^2)^\top$ and $\mathcal{C}_t = \mathcal{C}_{t-1} \cup \{t\}$

    Update $\pi_{t+1}(y|x) \propto \pi_t(y|x) \exp\left(\eta \widehat{D}_t(y, x)\right)$

  end if

end for

APPO — Three Key Components

(1) Regularised MLE estimator. $\hat{\theta}_t$ solves over queried rounds C_{t-1}:

$$\lambda\theta + \sum_{\tau \in C_{t-1}} \left[o_\tau - \sigma(\langle \theta, \phi_\tau^1 - \phi_\tau^2 \rangle)\right] (\phi_\tau^1 - \phi_\tau^2) = 0$$

Only queried data enters the estimator. Confidence ellipsoid: $\|\theta^* - \hat{\theta}_t\|_{\Sigma_{t-1}} \leq \beta$ with $\Sigma_t = \lambda I + \sum_{\tau \in C_t} (\phi_\tau^1 - \phi_\tau^2)(\phi_\tau^1 - \phi_\tau^2)^\top$.

(2) Optimistic reward gap (UCB). $\hat{D}_t(x, y) = \min\left\{\langle \hat{\theta}_t, \phi(x, y) - \phi_t^2 \rangle + \beta \|\phi(x, y) - \phi_t^2\|_{\Sigma_{t-1}^{-1}}, 1\right\}$ By Cauchy–Schwarz, $\hat{D}_t(x, y) \geq D_t(x, y)$ on the confidence event.

(3) Policy update (OPPO).

$$\pi_{t+1}(y \mid x) \propto \pi_t(y \mid x) \cdot \exp\big(\eta \hat{D}_t(y, x)\big)$$

Update *only* on queried rounds.

APPO — Uncertainty-Aware Query Criterion

Decision rule at round t:

Choose $y_t^1 = \arg\max_y \hat{D}_t(x_t, y)$,
$y_t^2 \sim \text{Uniform}(\mathcal{A})$.

$$\text{If } \underbrace{\|\phi_t^1 - \phi_t^2\|_{\Sigma_{t-1}^{-1}}}_{\text{uncertainty}} \leq \Gamma : \quad \text{skip query}$$

$$\text{If } \|\phi_t^1 - \phi_t^2\|_{\Sigma_{t-1}^{-1}} > \Gamma : \quad \text{query human, add } t \to C_t$$

Intuition: $\|\cdot\|_{\Sigma_{t-1}^{-1}}$ is large for directions not yet explored by the design matrix Σ_{t-1} — i.e., comparisons the estimator hasn't seen.

Why skipping is safe

When uncertainty $\leq \Gamma$, Lemma E.3 guarantees $2\beta\Gamma < \Delta$. Combined with the UCB optimism:

$$D_t(x_t, y_t^*) - D_t(x_t, y_t^1) \leq 2\beta\Gamma < \Delta$$

By Assumption 3.4 ($\Delta > 0$), this means y_t^1 *is* optimal — zero regret incurred.

Main Theorem and Comparison

Theorem 5.1

Set $\Gamma = \tilde{\mathcal{O}}(\Delta/\sqrt{d})$, $\lambda = B^{-2}$, $\eta = \tilde{\mathcal{O}}(\sqrt{\Gamma^2 \log |\mathcal{A}|/d})$, $\beta = \tilde{\mathcal{O}}(\sqrt{d}/\kappa_\sigma)$. Then with probability $\geq 1 - \delta$:

$$\mathrm{Regret}(T) = \tilde{\mathcal{O}}\left(\frac{d^2}{\Delta}\right), \qquad \mathrm{Query}(T) = |C_T| = \tilde{\mathcal{O}}\left(\frac{d^2}{\Delta^2}\right)$$

Method	**Regret**	**Query**	$\lvert\mathcal{A}\rvert$**-free?**
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Proof Sketch — Regret Decomposition

Decompose regret over queried vs. non-queried rounds:

$$\mathrm{Regret}(T) = \underbrace{\sum_{t \in C_T} \left[D_t(x_t, y_t^*) - D_t(x_t, y_t^1) \right]}_{I_1} + \underbrace{\sum_{t \notin C_T} \left[D_t(x_t, y_t^*) - D_t(x_t, y_t^1) \right]}_{I_2}$$

$I_2 = 0$ **(non-queried rounds):**
For $t \notin C_T$: $\|\phi_t^1 - \phi_t^2\|_{\Sigma_{t-1}^{-1}} \leq \Gamma$, so

$$D_t(x_t, y_t^*) - D_t(x_t, y_t^1) \leq 2\beta\Gamma < \Delta$$

By Assumption 3.4, y_t^1 is *optimal* $\Rightarrow$ contribution is exactly 0.

$I_1 = \tilde{\mathcal{O}}(d^2/\Delta)$ **(queried rounds):** Split further: $I_1 = J_1 + J_2 + J_3$ where

- J_1: UCB estimation error, bounded via Cauchy–Schwarz + elliptical potential lemma: $J_1 \leq \tilde{\mathcal{O}}(\beta\sqrt{|C_T| d})$
- J_2: OPPO sub-optimality via online mirror descent: $J_2 \leq \tilde{\mathcal{O}}(\sqrt{d\Gamma^{-2} \log |\mathcal{A}|})$
- J_3: Martingale noise (Azuma–Hoeffding): $J_3 \leq \tilde{\mathcal{O}}(\sqrt{|C_T|})$

Plug in Lemma E.1: $|C_T| = \tilde{\mathcal{O}}(d/\Gamma^2) \Rightarrow I_1 = \tilde{\mathcal{O}}(\beta d/\Gamma) = \tilde{\mathcal{O}}(d^2/\Delta)$.

ADPO — Adapting APPO to Neural DPO

Challenge: rewards are no longer linear; Σ_t^{-1} UCB bonus is unavailable.

Confidence estimator (replaces UCB bonus):

$$C_\theta(x, y^1, y^2) = |r_\theta(x, y^1) - r_\theta(x, y^2)|$$

where $r_\theta(x, y) = \beta[\log \pi_\theta(y|x) - \log \pi_{\text{ref}}(y|x)] + \beta Z(x)$.

$C_\theta \leq \gamma$: model is uncertain $\Rightarrow$ **query human**.
$C_\theta > \gamma$: model is confident $\Rightarrow$ **use pseudo-label**.

Training label (key design):

$$o_\theta(x, y^1, y^2) = \begin{cases} o & C_\theta \leq \gamma \text{ (hu} \\ \operatorname{sign}\big(r_\theta(x, y^1) - r_\theta(x, y^2)\big) & C_\theta > \gamma \end{cases}$$

Modified DPO loss:

$$\mathcal{L}_D = -\mathbb{E}\big[\log \sigma\big(o_\theta \cdot (r_\theta(x, y^1) - r_\theta(x, y^2))\big)\big]$$

Pseudo-labels keep high-confidence pairs *in the gradient computation* rather than discarding them.

Experiments

Setup

- Models: Zephyr (Beta, Gemma)
- Compare DPO vs ADPO

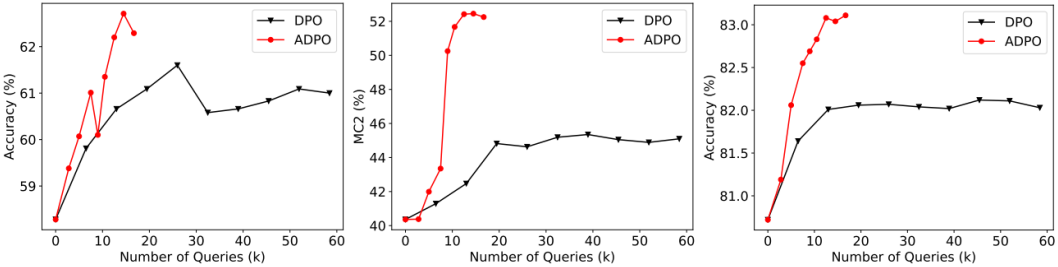
Key Results

- ADPO achieves similar or better performance than DPO
- Uses significantly fewer human queries (up to 50% reduction)

Insight

Querying only uncertain pairs reduces labeling cost without sacrificing performance

Experimental Results



(a) ARC Challenge (b) TruthfulQA (c) HellaSwag

Summary, Limitations, and Open Questions

Summary

- Formalises query-efficient RLHF as active linear contextual dueling bandits
- APPO: first $|\mathcal{A}|$-free instance-dependent bound; constant regret + finite query budget
- ADPO: pseudo-label-based practical algorithm; $\geq 50\%$ label reduction with equal or better performance
- **Key idea: query only uncertain pairs, and reuse confident pairs via pseudo-labels**

Limitations & open questions

- **Theory–practice gap:** guarantees only hold for linear APPO, not neural ADPO
- **Pseudo-label quality:** model may reinforce its own mistakes
- **Unknown Δ:** threshold selection requires prior knowledge or tuning

Core message

Querying selectively based on model uncertainty — and reinforcing confident predictions via pseudo-labels — reduces human labeling cost while maintaining performance.